

SIR OTONDE MICHAEL AWITI



KENYATTA UNIVERSITY

EXAMINATION FOR THE DEGREE OF BACHELOR OF EDUCATION

SMA 113: ANALYTICAL GEOMETRY

1ST SEMESTER 2023/2024

TIME: 2 HOURS

INSTRUCTIONS: Answer question ONE (Compulsory) and any other TWO questions

QUESTION ONE (30MARKS)

- Q1 (a) Find the equation of the line passing through the point A (2, -1) and perpendicular to the line joining the points B (4, 3) and C (-2, 5) (4marks)
- (b) Obtain the equation of the tangent to the circle
 $2x^2 + 2y^2 - 8x - 5y - 1 = 0$ at point (1, -1) (4marks)
- (c) Find the equation of a circle with radius 8 units and center at the intersection of $x + 2y = 8$ and $2x - 3y = 5$ (4marks)
- (d) Find the equation of the parabola given that the point (-3, 0) is the focus and the line $x + 5 = 0$ is the directrix. (4marks)
- (e) Find the equation of the ellipse with center at (-1, -1), vertex (5, -1) and eccentricity $= \frac{2}{3}$ (4marks)
- (f) Change $r^2 = 9 \cos 2\theta$ to rectangular equation form (4marks)
- (g) Evaluate the following without using tables or calculators
- (i) $2 \sin \frac{\pi}{8} \cos \frac{\pi}{8}$ (2marks)
- (ii) $2 \cos^2 15^\circ - 1$ (2marks)
- (iii) $\frac{1 - \sin^2 79^\circ}{1 - \cos^2 11^\circ}$ (2marks)

SIR OTONDE MICHAEL AWITI

SIR OTONDE MICHAEL AWITI

QUESTION TWO (15MARKS) ✓

- a) Derive the equation of a parabola with its vertex at point $(-2, 3)$ and focus at the point $(1,3)$.
(5marks)
- b) Express $3\cos x + 4\sin x$ in the form $r\cos(x + \alpha)$ where r and α are to be determined. Hence solve the equation $3\cos x + 4\sin x = -2$ for the domain $0^\circ \leq x \leq 360^\circ$ (10marks)

QUESTION THREE (15 MARKS) ✕

- (a) (i) Give the definition of a parabola (2 Marks)

- (ii) Find the focus and directrix of the parabola

$$x^2 = 14y \quad (3 \text{ Marks})$$

- (iii) Find the equation of the parabola having the vertex at $(0,0)$, axis of symmetry along the x - axis and passing $(2,-1)$ (3 Marks)

- (b) (i) Find the centre and radius of the circle

$$x^2 + y^2 + 8x + 6y = 0 \quad (3 \text{ Marks})$$

SIR OTONDE MICHAEL AWITI

- (ii) Find the points of intersection of the circle

$$x^2 + y^2 - x - 3y = 0 \quad (4 \text{ Marks})$$

And the line

$$y = x - 1$$

QUESTION FOUR (15 MARKS)

- (a) (i) Convert the polar coordinates $(4, \frac{\pi}{4})$ into rectangular coordinates. (3 Marks)
- (ii) Convert the following equation into polar form:

$$4x^2 + 9y^2 = 36 \quad (3 \text{ Marks})$$

SIR OTONDE MICHAEL AWITI

(b) (i) Prove the identity

$$\frac{\operatorname{cosec} \beta}{\sin \beta} - \frac{\cot \beta}{\tan \beta} = 1 \quad (3 \text{ Marks})$$

(ii) Solve the trigonometric equation

$$\tan^2 x - \tan x - 2 = 0 \quad [0, 2\pi] \quad (3 \text{ Marks})$$

(iii) Simplify $\frac{\cos x}{1 + \sin x} + \tan x$ (3 Marks)

QUESTION FIVE (15 MARKS) ✓

(a) (i) Give the definition of ellipse (2 Marks)

(ii) Find the coordinate of the vertices and foci of the ellipse

$$\frac{x^2}{100} + \frac{y^2}{64} = 1 \quad (5 \text{ Marks})$$

(b) (i) Give the definition of hyperbola. (2 Marks)

(ii) Determine and plot the coordinates of foci, vertices and eccentricity of the hyperbola

$$\frac{x^2}{144} - \frac{y^2}{81} = 1 \quad (6 \text{ Marks})$$

SIR OTONDE MICHAEL AWITI

SIR OTONDE MICHAEL AWITI

SIR OTONDE MICHAEL AWITI

DEFINITIONS AND FORMULAE

Area and volume formulae

Volume of a cone or pyramid	$= \frac{1}{3}Ah$, where A = base area, h = height of vertex.
Area of curved surface of a cone	$= \pi rl$, where l = slant height, r = base radius.
Volume of a sphere	$= \frac{4}{3}\pi r^3$.
Surface area of a sphere	$= 4\pi r^2$.
Area of a spherical zone (between planes distance h apart)	$= 2\pi rh$.

Trigonometry

$$\sec \theta = \frac{1}{\cos \theta}; \quad \csc \theta = \frac{1}{\sin \theta}; \quad \tan \theta = \frac{\sin \theta}{\cos \theta}; \quad \cot \theta = \frac{1}{\tan \theta}.$$

$$\cos^2 \theta + \sin^2 \theta = 1; \quad 1 + \tan^2 \theta = \sec^2 \theta; \quad \cot^2 \theta + 1 = \csc^2 \theta.$$

$$\sin(\theta \pm \phi) = \sin \theta \cos \phi \pm \cos \theta \sin \phi; \quad \cos(\theta \pm \phi) = \cos \theta \cos \phi \mp \sin \theta \sin \phi;$$

$$\tan(\theta \pm \phi) = \frac{\tan \theta \pm \tan \phi}{1 \mp \tan \theta \tan \phi} \quad [\theta \pm \phi \neq (k + \frac{1}{2})\pi].$$

$$\sin 2\theta = 2 \sin \theta \cos \theta; \quad \cos 2\theta = \cos^2 \theta - \sin^2 \theta; \quad \tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta} \quad [\theta \neq (\frac{1}{2}k + \frac{1}{4})\pi].$$

$$2 \cos^2 \theta = 1 + \cos 2\theta; \quad 2 \sin^2 \theta = 1 - \cos 2\theta.$$

$$\text{If } t = \tan \frac{1}{2}\theta, \text{ then } \sin \theta = \frac{2t}{1+t^2}; \quad \cos \theta = \frac{1-t^2}{1+t^2}; \quad \tan \theta = \frac{2t}{1-t^2}; \quad \frac{d\theta}{dt} = \frac{2}{1+t^2}.$$

$$2 \sin \theta \cos \phi = \sin(\theta + \phi) + \sin(\theta - \phi);$$

$$2 \cos \theta \cos \phi = \cos(\theta + \phi) + \cos(\theta - \phi);$$

$$2 \sin \theta \sin \phi = \cos(\theta - \phi) - \cos(\theta + \phi).$$

$$\sin \alpha + \sin \beta = 2 \sin \frac{1}{2}(\alpha + \beta) \cos \frac{1}{2}(\alpha - \beta); \quad \sin \alpha - \sin \beta = 2 \cos \frac{1}{2}(\alpha + \beta) \sin \frac{1}{2}(\alpha - \beta);$$

$$\cos \alpha + \cos \beta = 2 \cos \frac{1}{2}(\alpha + \beta) \cos \frac{1}{2}(\alpha - \beta); \quad \cos \alpha - \cos \beta = 2 \sin \frac{1}{2}(\alpha + \beta) \sin \frac{1}{2}(\beta - \alpha).$$

In the triangle ABC :

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R;$$

$$a^2 = b^2 + c^2 - 2bc \cos A, \text{ etc.};$$

$$\tan \frac{1}{2}A = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}, \text{ etc.}; \quad \text{area} = \sqrt{s(s-a)(s-b)(s-c)};$$

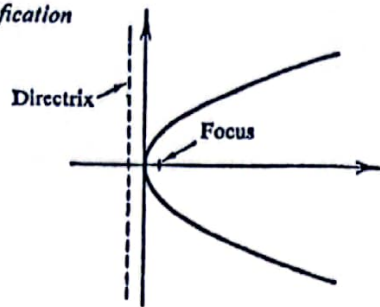
where $s = \frac{1}{2}(a+b+c)$.

Ranges of the inverse functions:

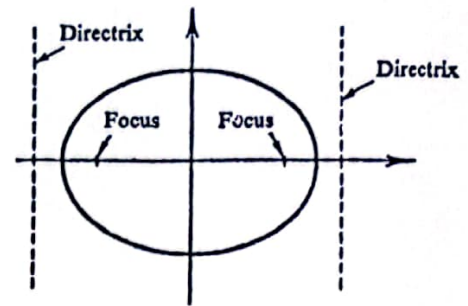
$$-\frac{1}{2}\pi \leq \sin^{-1}x \leq \frac{1}{2}\pi; \quad 0 \leq \cos^{-1}x \leq \pi; \quad -\frac{1}{2}\pi < \tan^{-1}x < \frac{1}{2}\pi.$$

DEFINITIONS AND FORMULAE

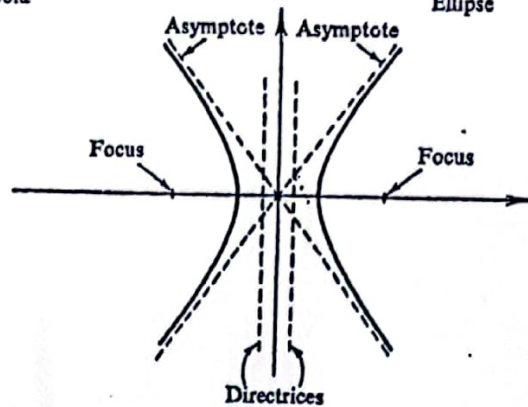
Conics
Classification



Parabola



Ellipse



Hyperbola

SIR OTONDE MICHAEL AWITI

Name of curve	Standard form of equation	Standard parametric forms	Eccentricity	Foci	Directrices	Asymptotes
Parabola	$y^2 = 4ax$	$(ap^2, 2ap)$	$e = 1$	$(a, 0)$	$x = -a$	None
Ellipse	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	$(a \cos \theta, b \sin \theta)$	$e = \frac{\sqrt{a^2 - b^2}}{a} < 1$	$(\pm ae, 0)$	$x = \pm \frac{a}{e}$	None
Hyperbola	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$(a \sec \phi, b \tan \phi)$ or $(\pm a \cosh u, b \sinh u)$	$e = \frac{\sqrt{a^2 + b^2}}{a} > 1$	$(\pm ae, 0)$	$x = \pm \frac{a}{e}$	$\frac{x}{a} \pm \frac{y}{b} = 0$

SIR OTONDE MICHAEL AWITI

SIR OTONDE MICHAEL AWITI